

Mensuration and Solids

Review Exercise Answers

Level-2

Miscellaneous

S1. Let the volume of the tank be V . Let the initial cross-sectional area and flow rate be A and f respectively. We have:

$$36fA = V$$

The new cross-sectional area and flow rate will be $(1.5)^2 A$ (since both the length and breadth have been increased by 50%) and $0.5f$. If the time taken to fill the tank now is T hours, we have:

$$(T)(2.25A)(0.5f) = V = 36fA$$

$$\Rightarrow T = 32$$

Thus, the tank will now fill up in 32 hours.

S2. The volumes added by pipes A and B in 10 minutes and 25 minutes (respectively) are:

$$\begin{aligned} V_1 &= (3\text{ cm})(4\text{ cm})(10\text{ cm/sec})(600\text{ sec}) \\ &= 72,000\text{ cm}^3 \\ V_2 &= (6\text{ cm})(3\text{ cm})(10\text{ cm/sec})(1500\text{ sec}) \\ &= 2,70,000\text{ cm}^3 \end{aligned}$$

Thus, the total volume added to the tank is

$$\begin{aligned} &72,000\text{ cm}^3 + 2,70,000\text{ cm}^3 \\ &= 3,42,000\text{ cm}^3 \end{aligned}$$

If the height of the water in the tank is h cm, we have:

$$\begin{aligned} \pi(120\text{ cm})^2(h\text{ cm}) &= 3,42,000\text{ cm}^3 \\ \Rightarrow h &\approx 7.5 \end{aligned}$$

Thus, the height of water in the tank is about 7.5 cm.

S3. (a) The amount of rainfall which falls daily on the field is 1800 m^3 . The volume of the hemispherical tank is roughly $32,723\text{ m}^3$ (verify these calculations). Let us see how much water is wasted everyday because of flow control. The maximum volume which can flow through one pipe is

$$\begin{aligned} &(10\text{ cm})(8\text{ cm})(20\text{ cm/sec}) \\ &= 1600\text{ cm}^3/\text{sec} \end{aligned}$$

Now, in 2 hours, the volume of water which 100 pipes can output is:

$$(1600\text{cm}^3 / \text{sec})(7200\text{sec})(100) \\ = 1152\text{m}^3$$

The volume of water which gets wasted is

$$1800\text{m}^3 - 1152\text{m}^3 = 648\text{m}^3$$

(b) The number of days required in this scenario to fill the tank completely is

$$\frac{32723\text{m}^3}{1152\text{m}^3} \approx 28.4$$

If daily conditions are the same, 29 days would be required to fill the tank completely.

(c) The flow rate must be increased by a factor of

$$\frac{1800}{1152} = 1.5625$$

Thus, the increased flow rate should be

$$1.5625 \times 20 = 31.25\text{ cm} / \text{sec}$$

(d) The number of days required now would be

$$\frac{32723\text{m}^3}{1800\text{m}^3} \approx 18.2$$

Thus, 19 days would be required.

S4. Self-exercise. The answer is INR 1.43 lacs.

S5. Self-exercise. The answer is INR 45 crores. Hints: The inside volume of the container (excluding all the partitions) comes out to $21,600\text{ cm}^3$. The volume taken up by the partitions is $3,600\text{ cm}^3$. Of the remaining volume, half is occupied by the liquid.

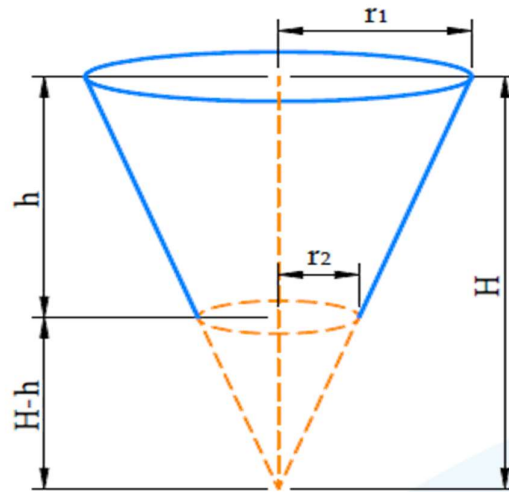
S6. We need to calculate the number of tea-sales possible with the two kinds of glasses. The inner radius of the tea vessel is 15 cm. The volume of tea which the vessel can store is:

$$\frac{2}{3}\pi(15)^3 \approx 7068.6\text{ cm}^3$$

Now, the inside volume of the cylindrical glass is:

$$V_{\text{cyl}} = \pi(2)^2(6.5) = 81.7\text{ cm}^3$$

Next, we need to calculate the inside volume of the frustum shaped glass. Consider the following figure, which shows a general frustum:



We will calculate the volume of this frustum in terms of the two base radii and its height h . The height of the cone of which this frustum is a part has been assumed as H . Now, the required volume is:

$$\begin{aligned} & \frac{1}{3}\pi r_1^2 H - \frac{1}{3}\pi r_2^2 (H-h) \\ &= \frac{1}{3}\pi (r_1^2 H - r_2^2 (H-h)) \end{aligned}$$

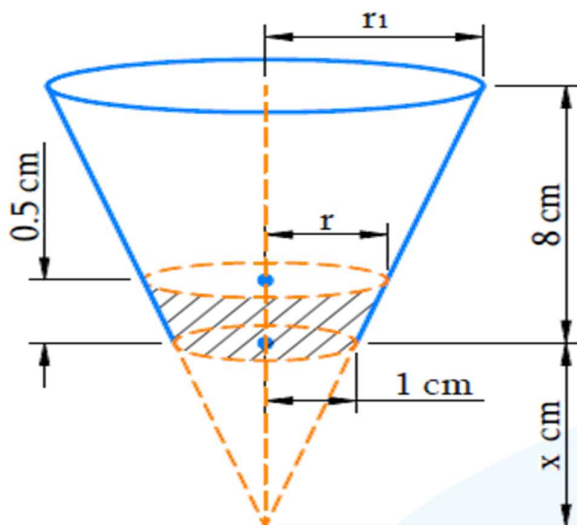
Also, using similarity, we have:

$$\begin{aligned} \frac{H-h}{H} &= \frac{r_2}{r_1} \\ \Rightarrow 1 - \frac{h}{H} &= \frac{r_2}{r_1} \Rightarrow H = \frac{hr_1}{r_1 - r_2} \end{aligned}$$

Thus, the volume is

$$\begin{aligned} V &= \frac{1}{3}\pi (r_1^2 H - r_2^2 (H-h)) \\ &= \frac{1}{3}\pi H \left(r_1^2 - r_2^2 \left(1 - \frac{h}{H} \right) \right) \\ &= \frac{1}{3}\pi \left(\frac{hr_1}{r_1 - r_2} \right) \left(r_1^2 - \frac{r_2^3}{r_1} \right) \\ &= \frac{1}{3}\pi h (r_1^2 + r_2^2 + r_1 r_2) \end{aligned}$$

Coming back to our frustum-shaped glass, consider the following figure now, which analyzes the geometry of this glass in detail:



We need to find the radius r of the inner base. Using similarity, we have:

$$\frac{x+8}{3} = \frac{x+0.5}{r} = \frac{x}{1}$$

$$\Rightarrow x = 4, r = \frac{4.5}{4} = 1.125$$

Now, we can calculate the inside volume of the frustum shaped glass:

$$V_{frus} = \frac{1}{3} \pi h (r_1^2 + r_2^2 + r_1 r_2)$$

$$= \frac{1}{3} \pi (7.5) (3^2 + 1.125^2 + 3 \times 1.125)$$

$$\approx 107.1 \text{ cm}^3$$

Now, we calculate the number of glasses which can be sold in the two cases:

$$n_{cyl} = \frac{7068.6 \text{ cm}^3}{81.7 \text{ cm}^3} \approx 86.5$$

$$n_{frus} = \frac{7068.6 \text{ cm}^3}{107.1 \text{ cm}^3} = 66$$

Finally, the money made in the two cases will be:

$$86 \times 6 = 516, 66 \times 6 = 396$$

The difference in the money made is Rs. 120.